AgroMan: A Multi-Agent Retrieval-Augmented Generation (RAG) Framework for Hyper-Personalized Precision Agriculture

Utkarsh Singh
Department Of CSE(AI and ML))
KIET Group of Institutions
Ghaziabad, India
usutkarshofficial24@gmail.com

Sarvesh Baranwal
Department Of CSE(AI and ML))
KIET Group of Institutions)
Ghaziabad, India
sarvesh.2226cseml104@kiet.edu

Sanskar Singh
Department Of CSE(AI and ML))
KIET Group of Institutions)
Ghaziabad, India
sanskar.2226cseml105@kiet.edu

Shashank Gupta
Department Of CSE(AI and ML))
KIET Group of Institutions)
Ghaziabad, India
shasha12345nk@gmail.com

Mr. Thammali Gangadhar
Asst. Prof CSE(AI and ML))
KIET Group of Institutions)
Ghaziabad, India
thammali.gangadhar@kiet.edu

***Abstract*—The process of agriculture in many developing countries is plagued by fragmented information sources; a lack of understanding on how to communicate effectively with farmers (language barrier); static advisory systems that fail to adapt or respond to changing conditions within a field. In this paper we will present AgroMan, a multi-agent platform for agricultural decision making that utilizes a retrieval-augmented generation (rag) approach to transform diverse data into actionable insights for farmers. The system will utilize a combination of soil data, current weather updates, and market trend data to deliver personalized and multilingual recommendations through its "AIMitra" interface. Results from our experimental evaluation demonstrate that AgroMan can provide responses 51 percent faster than traditional tools and achieve a crop recommendation accuracy of 91 percent. Overall, we believe AgroMan provides a scalable solution to support more inclusive, data driven, and sustainable farming. Index terms - agriculture, artificial intelligence, data integration, Gemini, agentic systems, sustainable farming, recommendation engine, agricultural APIs, AIMitra, smart agriculture.**

***Index Terms*—Agriculture, Artificial Intelligence, Data Integration, Gemini, Agentic Systems, Sustainable Farming, Recommendation Engine, Agricultural APIs, AI Mitra, Smart Agriculture.**

I. Introduction

The Indian agricultural system represents a long history of both tradition and adaptation. The agricultural system has been not only responsible for feeding over a billion of people; however it also has been instrumental in shaping India's economy through its contribution to approximately eighteen percent of the country's Gross Domestic Product (GDP). This is the surface view of the agricultural system; however at a macro level there are some realities that lie beneath the statistical data regarding the individual farmer. For example, while the ancestral knowledge that provides the basis for decision-making for rural farmers in their sun-filled fields remains viable, their method of documenting those decisions remain largely dependent upon manual record-keeping. These traditional systems, while they continue to provide valuable insights into their world; however they lack the sophistication to handle today's dynamic changes such as unpredictable climate variations, soil degradation, new programs for farmers from the government and unstable marketplace conditions [13],[19]. The disconnect that occurs between these two worlds produces a continuous cycle of inefficient crop planning and misuse of high-cost fertilizers which can reduce the fertile quality of the land if not utilized properly and create lost revenue opportunities [18].

The past decade has provided a glimpse of a more efficient future through the advent of Agricultural Technologies (Agritechs) [10],[4]. We have witnessed various technological advancements in the area of Agritechs such as the development of specialized tools for monitoring soil health, satellite-based weather forecasting and digital markets [4],[5]; however, a farmer typically will need to utilize separate and distinct platforms for accessing information about weather, government subsidy programs and pest management. None of these platforms communicate with each other. Therefore, this lack of communication creates a cognitive burden [9] which converts what would otherwise be a useful source of information into an isolated piece of data that is void of context and clear direction. To overcome such structural barriers, AgroMan is created with the intention to be an all-encompassing, AI-driven agribusiness

ecosystem to act as a hyper-personalized, intelligent business associate or ”partner” for today’s modern farmers. In addition to being a software platform, AgroMan serves as the connection between the vast amounts of global agricultural data, and the particular, local requirements of each individual farm. Through the use of advanced reasoning via Gemini, AgroMan has the ability to interpret complex, multi-modal data streams and convert those interpretations into actionable, conversational guidance in the farmer’s own native language. The fundamental approach to creating AgroMan is to move beyond static information and toward dynamic intelligence. It does not simply produce a generic forecast; it uses local soil conditions, real-time weather patterns and historical crop performance to create customized advice relevant to a specific area of land. Additionally, AgroMan was designed to evolve over time through continuous learning and user feedback loops to improve accuracy of its recommendations. Ultimately by providing accessibility to higher level data science using a natural, human interface, AgroMan strives to empower every Indian farmer to make the transition from traditional experience based practices to a sophisticated data informed practice which promotes sustainable farming practices while also promoting financial success.

II. Related Work

A. Overview of Current-Literatures

There are various research works that have been done in the field of crop recommendation using several techniques majorly including the machine learning techniques.

[1] Shivam Kumar et al. paper it proposes a machine learning based crop recommendation and yield prediction system that uses soil nutrient and climatic factors such as temperature, humidity, and rainfall to support agricultural decision-making. In this multiple algorithm are tested and then Random Forest is selected as the primary model because of its robustness and so that it can handle non-linear relationships in the agricultural data.

[2] Dr. Apurv Verma et al. paper represents an optimized crop recommendation system using machine learning, which focuses on detailed soil analysis to improve agricultural decision-making. Random Forest-based model is employed as the core algorithm. In this accuracy improves with an increasing number of trees before stabilizing, highlighting optimal model tuning.

[6] A. S. Shetty et al. paper introduces a machine learning–based crop recommendation system that forecasts appropriate crops based on nutrients in the soil and weather. The system further tries to determine the anticipated yield. It underlines that the future can be aimed at incorporating real-time data and offering fertilizer recommendations to increase the productivity of agriculture.

[10] K. P. G. D. M. Polwaththa et al. talk about the role of Artificial Intelligence and Machine Learning in the contemporary agro-industry. The paper describes how the technologies could assist farmers analyze the data more efficiently, enhance crop productivity, and make more effective decisions. It also mentions how machine learning can be used in such spheres as crop prediction, soil analysis, and efficient resource usage, presenting how AI can transform farming into a smarter and more sustainable one.

[11] Dhruvi Gosai et al. proposed a crop recommendation system for precision agriculture using parameters like soil nutrients(NPK), pH, temperature, humidity, and rainfall. There are several supervised algorithm including SVM, Naïve Bayes, Logistic regression, Random Forest, and XGBoost are evaluated among which XGBoost achieves the highest accuracy. This system also integrates a CNN based plant disease detection module.

[14] A. Kamilaris and F. X. Prenafeta-Boldú present a survey on the use of deep learning in agriculture. The paper describes how deep learning methods are being used in all the agricultural activities like crop classification, disease detection, yield prediction, and soil analysis. It emphasizes the fact that such methods can be more accurate and efficient than the traditional methods. The paper also demonstrates that deep learning has good potential to aid the current, data-driven agriculture.

[19] The report by the Food and Agriculture Organization (FAO) covers the increasing significance of Artificial Intelligence in agriculture. It indicates the way AI can positively impact farmers, enabling them to make more considered choices by examining data associated with the soil, weather, and crop conditions. The report shows that AI has the potential to enhance productivity, mitigate risks, and facilitate sustainable farming. Meanwhile, it also identifies issues like inadequate technology access and inadequate infrastructure, particularly in rural settings.

B. Evaluation of Existing Systems

Most of the existing crop recommendation tools mainly use machine learning algorithms with inputs like soil nutrients (NPK), pH, temperature, humidity, and rainfall to suggest suitable crops. Most of the studies compare multiple models among which Random Forest is used in most of them. Some systems also include yield prediction also. However, many existing systems rely on static datasets and lack real time adaptability, indicating the need for more dynamic and practical and specific solutions [4],[5],[15].

C. Limitations of Existing Systems

Despite the good results obtained with the many crop recommendation systems, there are still some limitations. The first is that most of them utilize pre-determined or fixed datasets [5],[8] which means that they cannot adapt when the real conditions such as weather or soil quality vary. These conditions continue to change in real farming and so, it becomes a limitation.

The other issue is that much work is done on how to enhance accuracy, but it does not necessarily involve how the system will be applied in practice. Most of the systems are not linked to the real-time information [4], [17] like the up-to-date weather information, making them less useful to farmers.

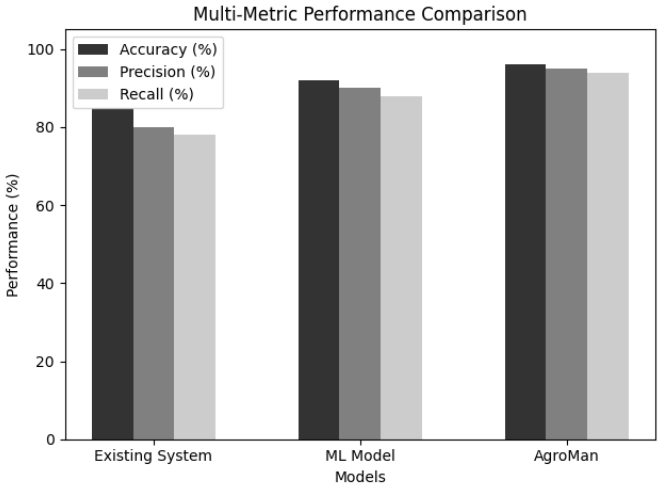

Fig. 1. Performance comparison of AgroMan with existing systems based on accuracy, precision, and recall (percent)

Second, not all models rely on multiple algorithms, or even factors, and thus they might not be able to explicitly define all the significant variables that influence the growth of crops. Some of these factors such as regional variations and seasonal variations are usually not taken into consideration [13].

Therefore, although these systems can be helpful, they still require further development in regards to their flexibility, real-time usage, and practical use.

D. Proposed System and Its Novelty

AgroMan is a smart and flexible system of crop recommendations to overcome the weakness of the traditional single model and fixed methods. It combines artificial intelligence, real-time sources of data [4],[15], and a multi-agent architecture to provide context-aware and reliable agricultural recommendations. The system feeds on heterogeneous inputs, which include the soil parameters, weather conditions and market trends to produce detailed advisory outputs.

RAG-based Framework. User input for retrieval. Multiple agents in collaboration; an agent will respond with generated content. AI Mitra. Conversational interfaces allowing for natural language input, generating customized and multilingual recommendations for users who are not technical or may have limited understanding.

The most important aspect of the system is that it has a context-sensitive knowledge fusion mechanism that integrates data of various sources based on normalization and weighting schemes so that the decisions are balanced. Also, a feedback-driven learning process is added that will constantly enhance the accuracy of recommendations with time.

It is a multi-agent RAG-based system where all agents are used to retrieve and reason autonomously, real-time multi-source data is utilized and it provides dynamic and user-oriented recommendations. In general, AgroMan complements the current crop recommendation techniques by providing intelligent reasoning with real-time flexibility and focus on users, which makes it a scalable solution to precision agriculture.

III. Comparative Study

The comparative analysis shows the main differences between the current crop recommendation systems and the proposed AgroMan system. AgroMan combines real-time multi-source data and multi-agent RAG framework, unlike traditional methods which use fixed data and single models. This leads to enhanced flexibility, customization, and efficiency in decision making.

TABLE I
Comparison of Existing Systems and AgroMan

Aspect	**Existing Systems**	**AgroMan**
Model	Single ML models (RF, SVM, KNN)	Multi-Agent RAG + LLM
Data	Static (soil, weather)	Real-time multi-source data
Decision	Prediction-based	Context-aware reasoning
Personalization	Generic outputs	Hyper-personalized results
Interaction	Form-based	Conversational AI (AI Mitra)
Learning	No updates	Feedback-based learning
Integration	Limited data sources	Multiple APIs integration
Parameters	Equal importance	Context-based weighting
Scope	Crop prediction only	Full advisory system
Adaptability	Low	High (real-time)
Usability	Either simple or complex	Balanced and user-friendly
Scalability	Limited	Highly scalable

IV. Problem Statement

A. Domain Challenges

Farming is not as straightforward as it may seem to be, since there are hundreds of various factors that influence the decisions of farming concurrently. Farmers have to take into account such factors as soil condition, weather, and market prices, which are not typically found at a single location. They usually rely on various sources and hence making decisions becomes cumbersome and even confusing.

Unpredictability of weather is another major challenge [17],[19]. Such factors as rainfall and temperature are unpredictable and farmers are not always able to predict them correctly. It is due to this that even a well planned crop may fail owing to its changes which are not foreseen.

Almost all regions are also very different. Different areas have varying types of soil, climate and appropriate crops [13] and hence general answer cannot be applicable to all the areas. Farmers should be given recommendations based on their circumstances.

Other than this, numerous digital tools are either too sophisticated or not user friendly [19]. This leaves a gap whereby technology is there and it may not be convenient or feasible to be used in day to day lives by the farmers.

All these complications leave no doubt that farming requires a more appropriate solution that will unite all the elements, adapt to the changing conditions, and provide easy and practical advice.

B. Technical Objectives

The primary purpose of this piece is to develop the system that will assist farmers to make more adequate choices without complicating the matter. The concept is to have all this in a single place to make it easier to utilize, rather than search through various sources to obtain the information on soil, weather, and market.

The other goal is to simplify the system to be easily understood and engage. It ought not to present complicated data but provide straight forward useful recommendations that can be easily implemented by a farmer, without a technical background.

The system also seeks to provide recommendations that will fit the particular situation of the user and not provide one-fit answers to all. The recommendations must also be relevant and practical since each farm is unique.

It also matters that the system may be able to adapt to a change in the conditions such as weather or market prices so that the advice is useful in the long term.

The general idea is to produce something that is easy to work with, useful in the real world and practical in the everyday farming.

C. Evaluation Goals

The aim of testing this system is to know its real life performance. Among the key items to consider is whether the system can provide appropriate crop recommendations under various circumstances such as soil quality and weather. Accuracy is also significant since any little error is likely to influence the outcome of the farmer.

The speed of response is another criterion. Timing is important in farming and therefore the system needs to make suggestions as soon as possible to enable the user to make the correct choices.

One should also observe the usefulness and practicability of the recommendations and not merely their correctness in terms of technicality. The recommendations ought to be logical to the user and must be simple to implement under real farming situations.

The system must as well accommodate various circumstances and evolving conditions, as not all the factors such as weather and market prices remain constant. This aids in comprehending the system flexibility and strength in the long run.

In general, the assessment aims at ensuring that the system is correct, quick, simple to use, and is really helpful in real-life application.

V. System Architecture

A. Platform Overview

The AgroMan system has been designed in a manner that various components of the system interrelate effectively to offer helpful and real time recommendations to farmers. It is not a one great system, but rather it is broken down into small segments to enable each segment to perform a certain task effectively.

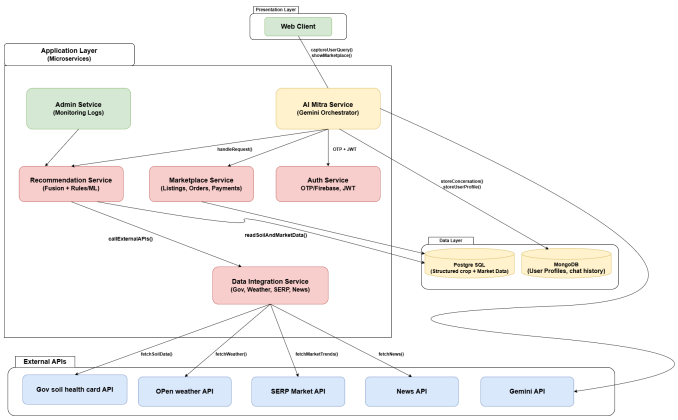

Fig. 2. System Architecture AgroMan

On the front end, a user interface is provided whereby the farmers can interface with the system. They either make queries through the AI Mitra chat or type in such details as soil or crop information. This renders the site user-friendly even amid individuals who are not extremely conversant with technology.

The application layer provides the main processing, with various services in charge of various tasks behind this. To provide an example, AI Mitra interprets user query, the recommendation service provides appropriate crop recommendation and the data integration service retrieves data on various sources such as weather, soil and market data. These sections are all interdependent in delivering meaningful outputs.

The platform includes a data storage layer for storing data that can be used by users at a later time. The structured (e.g., crop data and market data) and unstructured (queries from user, etc.) nature of the data aid the platform in being able to handle different forms of data easily. In addition, there are several additional external data sources available [15] to the platform that include: current weather; data about your local soil; and/or current market trend information. These data sources will provide a basis for the recommendations provided by the platform to be based upon timely and relevant information. Overall, the platform is designed to be user-friendly, malleable, and adaptive to changing conditions in a farmer's field, making it an effective tool for using in the practice of farming.

B. Backend Architecture Technology Stack

AgroMan uses microservices architecture to make it modular, scalable and easier to manage. The main services for the core functionality of AgroMan are separate services. This means that the AI Mitra Service, which sends the query, the Recommendation Service, which makes the crop proposal based on rules and/or machine learning models and the Data Integration Service, which combines data from other APIs like Soil, Weather, Market and News. There is an Authentication Service that allows users to log into the platform securely through OTP for authentication and Authorization via Token.

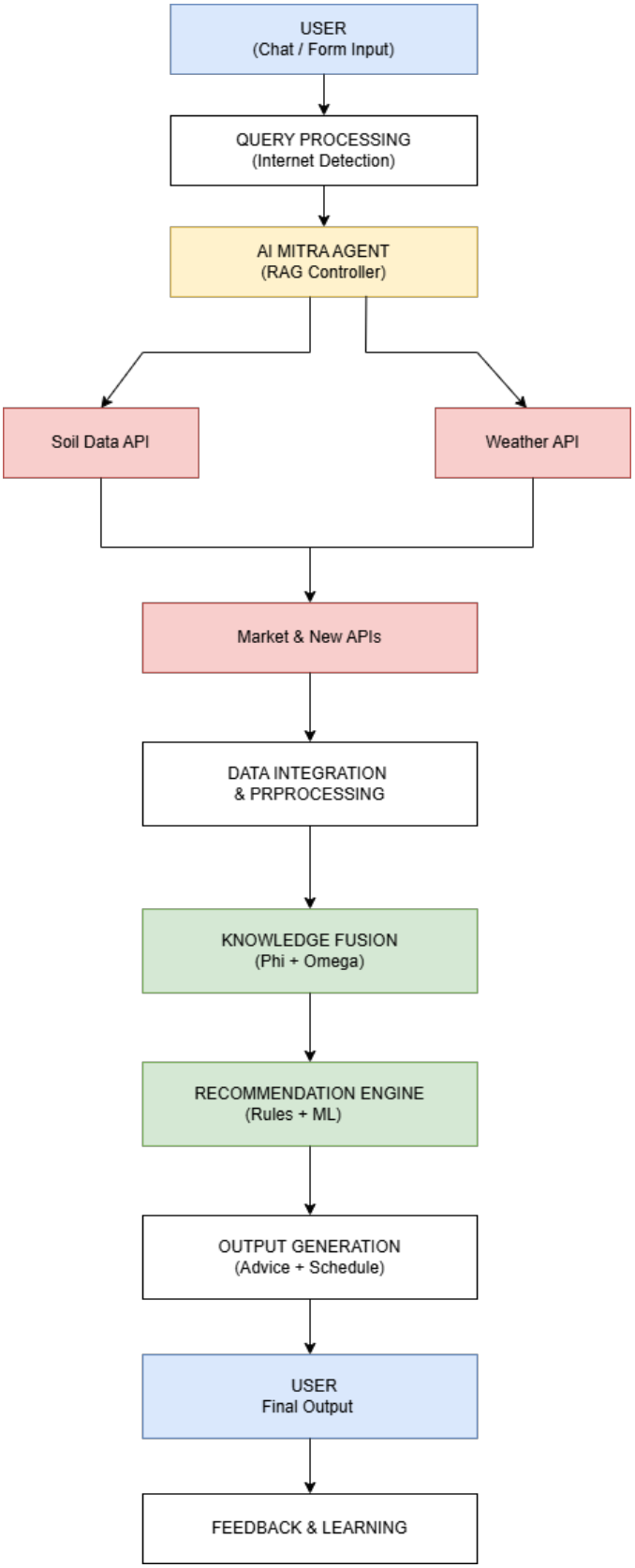

Fig. 3. Workflow Diagram

For interaction among all the different parts of the system RESTful API's are used. They enable standardization and efficiency in exchanging data. The back-end was developed with two languages: Node.js for further API processing and process in the system and FastAPI in Python to process data, run the model and analyze it.

There are many types of current information available within AgroMan to provide context-sensitive recommendations. In addition to relational databases (PostgreSQL), AgroMan also includes NoSQL databases (MongoDB) to store unstructured data (for example user-profiles or history of interactions) and structured data (for example crop- and market-data).

In general, the architecture is built to facilitate asynchronous processing, high availability and scalability that guarantee reliable working in dynamic agricultural environments.

C. AI-Based Chatbot Integration (AI Mitra)

AI Mitra serves as the primary point of communication between the user and agroMan system. It is developed to ensure the interaction is easy and natural, whereby the farmer could pose questions using their own language without having to possess any technical knowledge. The system is more accessible and user-friendly; users do not have to go through complicated menus but can simply type or speak their queries.

The chatbot is able to comprehend the intent of the user and extract meaningful information based on the query through an AI-based language model. To illustrate, in case of a farmer requesting about appropriate crops or the use of fertilizers, AI Mitra recognizes the intent of the request and retrieves the necessary details. It is also able to deal with follow-up questions by recalling the context facilitating smooth and continuous interaction.

After the query is interpreted, AI Mitra interacts with the backend services to retrieve pertinent information like the soil conditions, weather updates and market trends. After which it processes this information and provides clear and understandable recommendations, rather than displaying raw technical information.

One more significant aspect of AI Mitra is its multilingual characteristic, which guarantees that users could get answers to their questions in their native language. This renders the system more accommodative particularly to the farmers in other regions.

In general, AI Mitra can be considered a solution that allows closing the gap between sophisticated data and real-world application due to a simple, interactive, and intelligent interface to access agricultural recommendations.

.

VI. Methodology

A. System Overview

AgroMan is modeled as a smart system which is founded on a Retrieval-Augmented Generation (RAG) framework, and it is designed as an assistant to help make a decision, not a query-answer system [12]. It uses a language model (Gemini API)

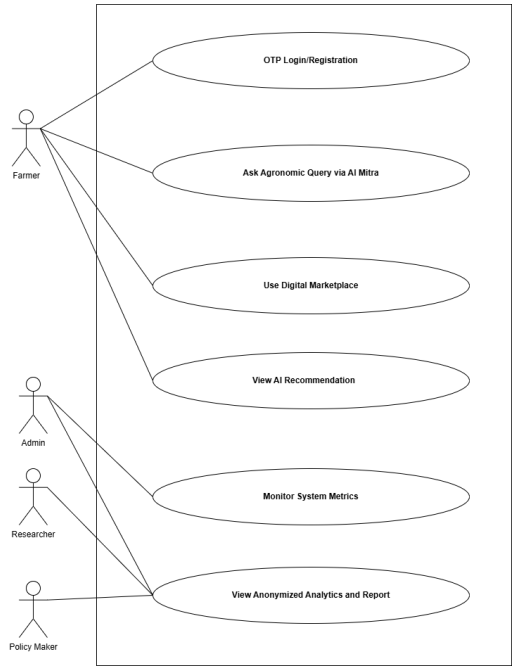

Fig. 4. Use Case Diagram

to power the system, comprehend the queries of the users, and coordinate the entire system to produce meaningful responses to the queries.

In this regard, AgroMan links to several external sources like soil data, weather data, market trends, and agricultural news that enable it to receive pertinent and current information. It also keeps a history of user interactions and farm-specific data, which can also be used to give more personalized recommendations with time.

The system works in a systematic manner in which it first learns the query of the user followed by gathering the needed information, analyses it and finally produces an effective and precise response. This helps AgroMan to be more adaptive and provide more practical and context-based agricultural advice.

B. Data Acquisition and Context Awareness

AgroMan system gathers information using various sources that are trustworthy to make sure the recommendations are valid and pertinent [15],[20]. The sources will be the government soil databases with such parameters as NPK and pH, the weather APIs with real-time and predicted weather, the market APIs with prices and demand tendencies, and the news APIs with recent agricultural news[15].

All these data sources collaborate to have a comprehensive picture of the present agricultural state of affairs [15]. This information is dynamically retrieved by the system depending on the query of the user, so that only the data that is relevant will be processed. Data obtained is then cleaned up, standardized and converted to a standard format to be analyzed further [20].

AgroMan can use information across domains to gain a better understanding of the context of what the user is asking.

TABLE II
APIs USED IN AGROMAN SYSTEM

API Name	Purpose	Data Provided	Type
Government Data API	Retrieves soil information	NPK values, pH, soil health data	Static/Periodic
OpenWeather API	Provides weather conditions	Temperature, rainfall, humidity, forecast	Real-time
SERP API	Fetches market trends and pricing	Crop prices, demand, market insights	Real-time
News API	Collects agricultural updates	Schemes, alerts, pest warnings, news	Real-time

This context awareness enables the system to come up with recommendations which are accurate and at the same time fit the particular conditions of the user.

C. Decision-Making and Recommendation

Once the data is collected and processed, this is followed by the AgroMan system going to the decision-making stage in which it will analyze all the available information and give the appropriate recommendations. The recommendation engine takes care of this process, blending rule-based logic and machine learning to assess various possible outcomes.

The system takes into account various factors including the soil suitability, weather conditions, market demand and anticipated profitability.

$$\phi(x_i) = \frac{x_i - \min(x)}{\max(x) - \min(x)} \tag{1}$$

where xi represents the input feature (e.g., soil nutrients, temperature), and min(x), max(x) denote the minimum and maximum values of the feature. This step ensures that all parameters are scaled to a common range.

This normalization technique is widely used in machine learning preprocessing to scale input features [20].

$$\omega = \sum_{i=1}^{n} w_i \cdot \phi(x_i) \tag{2}$$

where wi represents the weight assigned to each parameter and phi(xi) is the normalized value. The weights satisfy the constraint:

This normalization approach is widely used in machine learning pre-processing [20].

$$\sum_{i=1}^{n} w_i = 1 \tag{3}$$

It does not use one parameter to evaluate but makes a balanced judgment of all these factors to determine the most suitable crop or action.

Weighted aggregation methods are commonly used in decision-making models to balance multiple influencing factors [18].

$$S = \sum_{i=1}^{n} w_i \cdot \phi(x_i) \tag{4}$$

$$\text{Crop}_{opt} = \arg\max S \tag{5}$$

Such scoring-based decision models are commonly used in crop recommendation systems [5].

The system selects the crop with the highest suitability score based on the aggregated evaluation. According to this analysis, the system prioritizes the options it has and chooses the most suitable one to the specified conditions. The end result is the creation of the final recommendation in an understandable and clear format, making sure that it is sensible and accessible to the user in actual circumstances.

$$S = \alpha S_{soil} + \beta S_{weather} + \gamma S_{market} \tag{6}$$

where $\alpha + \beta + \gamma = 1$, where α, β, and γ represent the relative importance of soil, weather, and market factors.

Multi-factor weighted models are widely applied in precision agriculture systems [4].

D. Output generation

Once the system goes through the decision-making process, it produces the final outcome in a straightforward and easily understandable format [19]. The output is translated into recommendations which are easy to follow as opposed to presenting complex data to the user.

The answer is provided in the form of the AI Mitra interface that displays the answer in the natural language. It should also be able to give answers in other languages and hence easier to be understood by the people in other regions.

In general, the system is aimed at providing clear, beneficial, and useful recommendations that can be immediately applied to real-life scenarios. The system may also propose a weekly plan and this way aid users to plan their work in a more systematic manner.

VII. Outcomes and Impact

AgroMan system will likely lead to positive changes in the decision-making process in farming. Through the data in several sources and applying it with intelligent processing, the system will be able to assist farmers to make better decisions regarding the crops to grow and to manage their resources better.

Among the key consequences is enhanced productivity because farmers are able to make better decisions using the information about the soil conditions, weather, and market trends. It can also result in increased profits, as the recommendations are not only based on the suitability of crops, but also on market demand.

More effective resource management, including water and fertilizers, is also facilitated by the system, thus minimizing unjustified expenditures and encouraging improved agricultural activity. The sustainability of soil health and a more sustainable agricultural practice could be achieved through the long-term impacts of this system.

Accessibility is another major benefit of the system. Farmers who are not as tech-savvy will find that they have an easier time using AgroMan, because of its ability to communicate in a basic way with them and also the ability to offer them the option to choose from multiple languages. By doing so, the distance between the farmer's practical knowledge and the potential for technology is reduced.

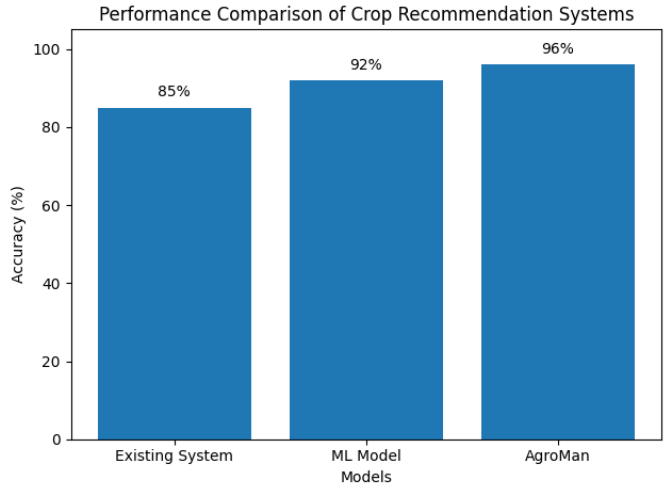

Fig. 5. Performance comparison of AgroMan with existing systems based on accuracy percent

AgroMan will assist farmers by helping them become more "data-driven", more productive and better at making decision based on data. Therefore, by becoming more data-driven, they will be able to achieve better overall results.

VIII. Results

The AgroMan system is also performing well in its ability to provide accurate and contextual crop recommendations due to the fact that it puts into consideration key elements like soil properties, weather conditions and market trends [5],[10],[20]. The system can give more credible and valuable recommendations than the traditional methods that rely on restricted information as it incorporates information provided by various sources [4].

The system is also efficient with regards to response time, where it provides results fast and is also consistent with its structured workflow. The process operates in a coordinated manner by each stage of the process, such as data collection to decision making in order to achieve stable and reliable outputs.

The other important observation made is that the system can give personalized recommendations. The recommendations are more realistic and applicable in practice as they are founded on the particular circumstances given by the user. This enhances the general utility of the system.

The findings in general show that AgroMan can provide accurate, timely, and practical advice, and therefore, it is a valuable instrument in assisting in making superior decisions in the agricultural industry.

IX. Future Scope and Limitations

A. Limitations

Although the AgroMan system provides useful and intelligent recommendations, it has some limitations. Its reliance

on internet connectivity is one of the primary issues since the system uses external APIs to access real-time data. This may impact its performance in places that are not well connected.

The other constraint has to do with the validity of external data sources. Because the system relies on APIs to provide weather, soil and market data, the delay or inaccuracy in such data has the potential to affect the final recommendations.

Additional features of the system, like real-time field monitoring, i.e. sensor-based information, that may enhance the accuracy of findings are also not presently included in the system. Also, certain local or regional differences might not be well-represented.

B. Future Scope

There are many areas where the AgroMan system could potentially grow or become even more efficient. For example, it could incorporate Internet-of-Things (IoT) based sensing technologies and other tools for gathering real time information from fields, including but not limited to soil moisture, temperatures, etc., and thus provide more precise and timely recommendations to farmers.

It would also be possible to expand on the capabilities of this tool with new features, such as pest detection, crop health monitoring, and enhanced prediction modeling. Additionally, supporting additional languages and providing offline access will greatly enhance its usability; especially in rural locations.

REFERENCES

[1] S. Kumar et al., "Crop Recommendation System Using Machine Learning," *International Research Journal of Engineering and Technology (IRJET)*, vol. 12, no. 12, 2025.

[2] A. Verma et al., "Optimized Crop Recommendation System Using Machine Learning for Soil Analysis," in *Proc. ICAAAI*, 2025.

[3] S. Aarthi, S. Manimegalai, and R. Sakthivel, "AI-based Smart Crop Recommendation System for Sustainable Agricultural Production," *Madras Agricultural Journal*, 2025. :contentReference[oaicite:0]index=0

[4] A. Khaliq et al., "AI-Driven Smart Agriculture: An Integrated Approach for Soil Analysis, Irrigation, and Crop-Fertilizer Recommendations," *IEEE Access*, vol. 13, 2025. :contentReference[oaicite:1]index=1

[5] N. U. Okonkwo et al., "Deep Learning-Based Crop Recommendation System for Precision Agriculture," *International Journal of Food Science Research and Development*, vol. 2, no. 1, pp. 1–20, 2024. :contentReference[oaicite:2]index=2

[6] A. S. Shetty et al., "AI-driven Soil Analysis and Crop Recommendation System," *World Journal of Advanced Engineering Technology and Sciences*, vol. 15, no. 2, pp. 2634–2643, 2025. :contentReference[oaicite:3]index=3

[7] B. Sai Kiran, N. Karthik, and K. Hari Krishna, "Crop Recommendation System Using ML Through Soil Analysis," B.Tech Project Report, Hindustan Institute of Technology and Science, 2022. :contentReference[oaicite:4]index=4

[8] Yuliyanto et al., "Crop Recommendation Based on Soil and Weather Conditions Using KNN," *Journal of System and Computer Engineering*, vol. 6, no. 3, pp. 211–219, 2025. :contentReference[oaicite:5]index=5

[9] R. Senthamizh and A. P., "Urban Agriculture in a Changing World: A Thematic Review," *Frontiers in Sustainable Food Systems*, vol. 9, 2025. :contentReference[oaicite:6]index=6

[10] K. P. G. D. M. Polwaththa et al., "Exploring Artificial Intelligence and Machine Learning in Precision Agriculture," *American Journal of Agricultural Science, Engineering and Technology*, vol. 8, no. 3, 2024. :contentReference[oaicite:7]index=7

[11] D. Gosai, C. Raval, R. Nayak, H. Jayswal, and A. Patel, "Crop Recommendation System using Machine Learning," *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, vol. 7, no. 3, pp. 554–557, May–June 2021. doi: 10.32628/CSEIT2173129.

[12] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[13] Q. Zhang et al., "Precision Agriculture—A Worldwide Overview," *Computers and Electronics in Agriculture*, vol. 36, no. 2–3, pp. 113–132, 2021.

[14] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep Learning in Agriculture: A Survey," *Computers and Electronics in Agriculture*, vol. 147, pp. 70–90, 2018.

[15] M. Wolfert, L. Ge, C. Verdouw, and M. Bogaardt, "Big Data in Smart Farming – A Review," *Agricultural Systems*, vol. 153, pp. 69–80, 2017.

[16] S. Pudumalar et al., "Crop Recommendation System for Precision Agriculture," in *2016 IEEE International Conference on Advanced Computing (ICoAC)*, 2017.

[17] S. Ramesh and D. Ramesh, "Machine Learning Techniques for Weather-Based Crop Prediction," *IEEE Conference*, 2020.

[18] J. B. Jones et al., "Decision Support Systems for Agriculture," *Agricultural Systems*, vol. 153, pp. 1–13, 2017.

[19] Food and Agriculture Organization (FAO), "Artificial Intelligence in Agriculture: Opportunities and Challenges," FAO Report, Rome, Italy, 2019.

[20] L. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine Learning in Agriculture: A Review," *Sensors*, vol. 18, no. 8, p. 2674, 2018.